

RPC Test Coverage by Node Type — Archive vs Full vs Snap

Ethereum mainnet JSON-RPC fixtures | current working tree vs origin/rpc-changes (27d4f79) | generated 2026-06-04

1. Executive summary

468

MAINNET ETH_ TESTS
(CURRENT)

175

TAGGED @ARCHIVE
(37.4%)

85

TRULY ARCHIVE-ONLY
(STATE)

90

@ARCHIVE BUT FULL-
NODE OK

Headline: Of 468 mainnet eth_* tests, **175** are tagged @archive. But only **85** of those genuinely need an **archive node** (they read *historical state*); the other **90** only need *historical chain data* (old blocks / receipts / logs), which a **full node retains**. So:

- Archive node:** all **468** pass.
- Full node:** **85 fail** (18.2%) — the historical-state tests. 383 pass.
- Snap-synced (default, full history backfilled):** same as full — **85 fail** (18.2%).
- Snap-synced (no-history / bodies+receipts download off):** **175 fail** (37.4%) — loses historical state *and* historical blocks/receipts.

2. origin/rpc-changes → current: what changed

Aspect	origin (27d4f79)	current	delta
Total mainnet eth_ tests	461	468	+7
Tests tagged @archive	0	175	+175
eth_getCode tests	8	14	+6
eth_getBlockByNumber tests	17	18	+1

Interpretation: origin/rpc-changes ships the @archive tagging *mechanism* (the scripts/tag_archive.sh script + runner support) but carries **no tags in the fixtures**. The current tree applies the tags to **175** fixtures, adds **7** new tests, and converts several historical-block tests in eth_call / eth_createAccessList / eth_simulateV1 to run at latest (so they no longer need archive). No tests were removed.

3. Coverage by request class

Every method maps to one class that determines node survivability.

Class	What it reads	Tests	@archive	Archive	Full	Snap (def)	Snap (no-hist)
STATE	historical account/storage state	253	85	all	-85	-85	-85
HISTORY	old blocks / txs / receipts / logs	192	90	all	all	all	-90
TIP	chain tip / stateless	23	0	all	all	all	all
TOTAL		468	175	468	383	383	293

4. Node capability matrix

Data retained	Archive	Full	Snap (default)	Snap (no-history)
Chain tip + recent state (~128 blk)	✓	✓	✓	✓
All blocks / txs / receipts / logs	✓	✓	✓	✗
Historical state (any block)	✓	✗	✗	✗
mainnet eth_ tests served	468	383	383	293

5. Per-method breakdown

Columns **FullX** / **SnapX** = tests that **fail** on that node type. STATE@archive fail everywhere except archive; HISTORY@archive fail only on a no-history snap node.

Method	Class	Total	origin	@archive	Full X	Snap X (def)	Snap X (no-hist)
eth_baseFee	TIP	1	1	0	0	0	0
eth_blobBaseFee	TIP	1	1	0	0	0	0
eth_blockNumber	TIP	1	1	0	0	0	0
eth_call	STATE	42	42	1	1	1	1
eth_callBundle	STATE	15	15	14	14	14	14
eth_callMany	STATE	15	15	15	15	15	15
eth_chainId	TIP	6	6	0	0	0	0
eth_coinbase	TIP	1	1	0	0	0	0
eth_createAccessList	STATE	22	22	10	10	10	10
eth_estimateGas	STATE	35	35	0	0	0	0
eth_feeHistory	HISTORY	22	22	19	0	0	19
eth_gasPrice	TIP	1	1	0	0	0	0
eth_getBalance	STATE	28	28	22	22	22	22
eth_getBlockByHash	HISTORY	10	10	0	0	0	0
eth_getBlockByNumber	HISTORY	18	17	9	0	0	9
eth_getBlockReceipts	HISTORY	10	10	7	0	0	7
eth_getBlockTransactionCountByHash	HISTORY	5	5	0	0	0	0
eth_getBlockTransactionCountByNumber	HISTORY	25	25	22	0	0	22
eth_getCode	STATE	14	8	7	7	7	7
eth_getFilterChanges	TIP	1	1	0	0	0	0
eth_getFilterLogs	TIP	1	1	0	0	0	0
eth_getLogs	HISTORY	13	13	13	0	0	13
eth_getProof	STATE	26	26	6	6	6	6
eth_getRawTransactionByBlockHashAndIndex	HISTORY	10	10	0	0	0	0
eth_getRawTransactionByBlockNumberAndIndex	HISTORY	13	13	9	0	0	9
eth_getRawTransactionByHash	HISTORY	16	16	0	0	0	0
eth_getStorageAt	STATE	10	10	3	3	3	3
eth_getTransactionByBlockHashAndIndex	HISTORY	6	6	0	0	0	0
eth_getTransactionByBlockNumberAndIndex	HISTORY	10	10	5	0	0	5
eth_getTransactionByHash	HISTORY	7	7	0	0	0	0
eth_getTransactionCount	STATE	8	8	2	2	2	2
eth_getTransactionReceipt	HISTORY	10	10	0	0	0	0
eth_getUncleByBlockHashAndIndex	HISTORY	3	3	0	0	0	0
eth_getUncleByBlockNumberAndIndex	HISTORY	5	5	2	0	0	2
eth_getUncleCountByBlockHash	HISTORY	3	3	0	0	0	0
eth_getUncleCountByBlockNumber	HISTORY	6	6	4	0	0	4
eth_getWork	TIP	1	1	0	0	0	0

eth_maxPriorityFeePerGas	TIP	1	1	0	0	0	0
eth_mining	TIP	1	1	0	0	0	0
eth_protocolVersion	TIP	1	1	0	0	0	0
eth_sendRawTransaction	TIP	3	3	0	0	0	0
eth_simulateV1	STATE	38	38	5	5	5	5
eth_submitHashrate	TIP	1	1	0	0	0	0
eth_submitWork	TIP	1	1	0	0	0	0
eth_syncing	TIP	1	1	0	0	0	0
TOTAL		468	461	175	85	85	175

6. Methodology & caveats

Scope: Ethereum **mainnet** only (integration/mainnet/eth_*); other networks carry tests but are not @archive-tagged.

@archive tag: applied by scripts/tag_archive.sh to any fixture whose block parameter is a fixed historical block (not latest/pending/earliest/safe/finalized). The script tags both state-reading and history-reading methods, so the raw tag count (175) over-states the true archive requirement.

STATE vs HISTORY split: STATE methods (eth_call, eth_estimateGas, eth_createAccessList, eth_simulateV1, eth_callMany, eth_callBundle, eth_getBalance, eth_getCode, eth_getStorageAt, eth_getProof, eth_getTransactionCount) read historical *state* → require an archive node. HISTORY methods (block/tx/receipt/log/feeHistory getters) read historical *chain data*, which a full node keeps → they do **not** need archive.

Full vs Snap: a fully-synced full node and a default snap-synced node both keep complete chain history but prune state to ~128 recent blocks, so they behave identically here. A snap node configured without body/receipt backfill (light/"no-history" mode) additionally loses the 90 HISTORY@archive tests.

eth_estimateGas: 0 archive tests — all its fixtures run at recent/latest blocks, so it is fully served by every node type.